

Multiple Choice (10 marks)

1. Which of the following is not a function of vitamin D?
 - (a) Lipid-soluble antioxidant in cell membranes
 - (b) Fast responses to increase calcium absorption with no change in gene expression
 - (c) Regulation of differentiation of adipocytes
 - (d) Regulation of bone turnover
2. The respiration quotient (RQ):
 - (a) is defined as the ratio of VO_2 divided by VCO_2 .
 - (b) decreases on a high carbohydrate diet.
 - (c) increases with fasting.
 - (d) goes beyond a value of 1.0 when exogenous carbohydrate is converted to endogenous fat.
3. Which of the following is a function of vitamin B₆?
 - (a) Synthesis of TMP (thymidine monophosphate)
 - (b) Decarboxylation of amino acids to form amine neurotransmitters
 - (c) Synthesis of methionine from homocysteine
 - (d) Carboxylation of pyruvate to oxaloacetate
4. In which form is selenium found in the 25 human selenoproteins?
 - (a) Selenophosphate
 - (b) Selenocysteine
 - (c) Selenohistidine
 - (d) Selenate
5. What mechanisms potentially accelerate age related cognitive decline (as of 2020 medical knowledge)
 - (a) Glucose neurotoxicity
 - (b) Insulin insensitivity
 - (c) Systemic inflammation
 - (d) All of the above

True / False (10 marks)

Answer True or False

6. True or False: On average, obese people have a higher basal metabolic rate due to having more lean mass compared to lean individuals.
7. True or False: Biotin is essential for the carboxylation of pyruvate to oxaloacetate in metabolic processes.
8. True or False: Underwater weighing is the most direct method for measuring body composition.
9. True or False: If the incidence rate of a disease decreases, the prevalence of the disease will also decrease in the population.
10. True or False: Prenatal programming only affects children who are born with low birth weights.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of various proteins in the absorption of cholesterol in the small intestine, highlighting which protein does not participate in this process and why.
12. Discuss the factors that influence the diagnosis of osteopenia and osteoporosis in older women, particularly those over age 80, and analyze the implications of bone mineral density in this population.

End of Paper